

Question

1. Write a Java program to check whether the given number is prime palindrome or not. (CO1)
2. Write a Java program which reads marks of five subjects through command line and prints the total and average.

Steps for Execution

1. Open your Java IDE (Eclipse/VS Code) or command prompt.
2. Create a new Java class file (PrimePalindrome.java and MarksCalculator.java).
3. Write the program code inside the class.
4. Compile the program using `javac FileName.java`.
5. Run the program using `java FileName`.
 - For Program 1: Enter a number when prompted.
 - For Program 2: Pass 5 marks as command line arguments.
6. Observe the output displayed on the console.

◆ Algorithm

Program 1: Prime Palindrome

1. Start.
2. Read a number from the user.
3. Check if the number is prime:
 - Divide by numbers from 2 to $n/2$.
 - If divisible, not prime.
4. Check if the number is palindrome:
 - Reverse the digits.
 - Compare with original.
5. If both prime and palindrome → print "Prime Palindrome".
6. Else → print "Not Prime Palindrome".
7. Stop.

Program 2: Marks Calculator

1. Start.
2. Read 5 subject marks from command line arguments.
3. Convert each argument to integer.
4. Add all marks to get total.
5. Divide total by 5 to get average.
6. Print total and average.
7. Stop.

UML Class Diagram

Program 1: PrimePalindrome

```

+-----+
| PrimePalindrome |
+-----+
| + isPrime(num)  |
| + isPalindrome(num)|
+-----+
| + main(args)   |
+-----+

```

Program 2: MarksCalculator

```

+-----+
| MarksCalculator |
+-----+
| + main(args)   |
+-----+

```

Code

Program 1: Prime Palindrome

```
import java.util.Scanner;
```

```
public class PrimePalindrome {

    public static boolean isPrime(int num) {
        if (num <= 1) return false;
        for (int i = 2; i <= num/2; i++) {
            if (num % i == 0) return false;
        }
        return true;
    }

    public static boolean isPalindrome(int num) {
        int original = num, reversed = 0;
        while (num > 0) {
            int digit = num % 10;
            reversed = reversed * 10 + digit;
            num /= 10;
        }
        return original == reversed;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int n = sc.nextInt();

        if (isPrime(n) && isPalindrome(n)) {
            System.out.println(n + " is a Prime Palindrome.");
        } else {
```

```
        System.out.println(n + " is NOT a Prime Palindrome.");
    }
    sc.close();
}
}
```

Program 2: Marks Calculator

```
import java.util.Scanner;

class MarksCalculator {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int total = 0;
        for (int i = 1; i <= 5; i++) {
            System.out.print("Enter marks of subject " + i + ": ");
            int marks = sc.nextInt();
            total = total + marks;
        }

        double average = (double) total / 5;

        System.out.println("Total Marks = " + total);
        System.out.println("Average Marks = " + average);
    }
}
```

Output Screenshots (Description)

Program 1: Prime Palindrome

Enter a number: 131

131 is a Prime Palindrome.

Program 2: Marks Calculator

Enter marks of subject 1: 80

Enter marks of subject 2: 75

Enter marks of subject 3: 90

Enter marks of subject 4: 85

Enter marks of subject 5: 70

Total Marks = 400

Average Marks = 80.0